Data Analytics Project: IGN Game Reviews Analysis

Dataset Used: ign.csv

This document outlines a structured, end-to-end data analysis project based on game reviews from the IGN dataset. The project is divided into five scenarios, increasing in difficulty from basic data handling to advanced analysis and visualization, ensuring comprehensive skill coverage.

Project Goal: To analyze game review scores, platform popularity, and genre distribution over time using the ign.csv dataset.

The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth (np.diff).
- **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts).

🎯 Project Coverage

Category	Skills Covered
Data Preprocessing	Data Loading, Missing Value Handling (Mean/Mode), Type Conversion, Column Removal
Pandas Operations	Grouping, Filtering, Aggregation, Feature Engineering (score_category, editors_choice conversion)
NumPy Calculations	Array conversion, Yearly score growth calculation (np.diff)
Visualization	Line Chart, Bar Chart, Pie Chart, Histogram, Stacked Bar Chart, Saving Plots

Scenario 1: Basic Data Loading & Cleaning (□ Easy)

Description:

In this scenario,

The dataset is loaded using Pandas. Basic checks like head, tail, and shape are performed. The unnecessary column is removed, missing values are handled (mean for score, mode for genre), and data types are corrected.

Task	Description
1.	Load the ign.csv dataset using Pandas.
2.	Display the first 5 rows (head()), last 5 rows (tail()), and the shape of the dataset.
3.	Remove the unnecessary column: "Unnamed: 0".
4.	Check for missing values in score, genre, and platform.
5.	Handle missing values: Fill the numeric column score with the mean, and the categorical column genre with the mode.
6.	Ensure correct data types: Convert score → float and release_year, release_month, release_day → integer.

Results of Scenario 1:

```
   Unnamed: 0 score_phrase                                      title  ... release_year release_month  release_day
0           0      Amazing                            LittleBigPlanet PS Vita  ...         2012            9           12
1           1      Amazing  LittleBigPlanet PS Vita -- Marvel Super Hero E...  ...         2012            9           12
2           2        Great                               Splice: Tree of Life  ...         2012            9           12
3           3        Great                                             NHL 13  ...         2012            9           11
4           4        Great                                             NHL 13  ...         2012            9           11

[5 rows x 11 columns]
Data Loaded Successfully
```

Fig-1.1 Displaying first 5 rows

```
       score_phrase                                      title  ... release_month release_day
18620          Good                  Tokyo Mirage Sessions #FE  ...             6          29
18621       Amazing          LEGO Star Wars: The Force Awakens  ...             6          29
18622      Mediocre  Star Ocean: Integrity and Faithlessness  ...             6          28
18623  Masterpiece                                     Inside  ...             6          28
18624  Masterpiece                                     Inside  ...             6          28

[5 rows x 10 columns]
Data Loaded Successfully
```

Fig-1.2 Displaying last 5 rows

Conclusion:

The dataset is now clean, consistent, and ready for analysis. Handling missing values and correcting data types ensures that future calculations and visualizations are accurate and reliable.

Scenario 2: Line Graph (Score Trend) + Save (□ Easy)

Description:

In this scenario,

The dataset is grouped by release year to calculate the average game score for each year. This helps identify how game quality has changed over time.

The grouped data is converted into NumPy arrays and plotted as a line graph, where:

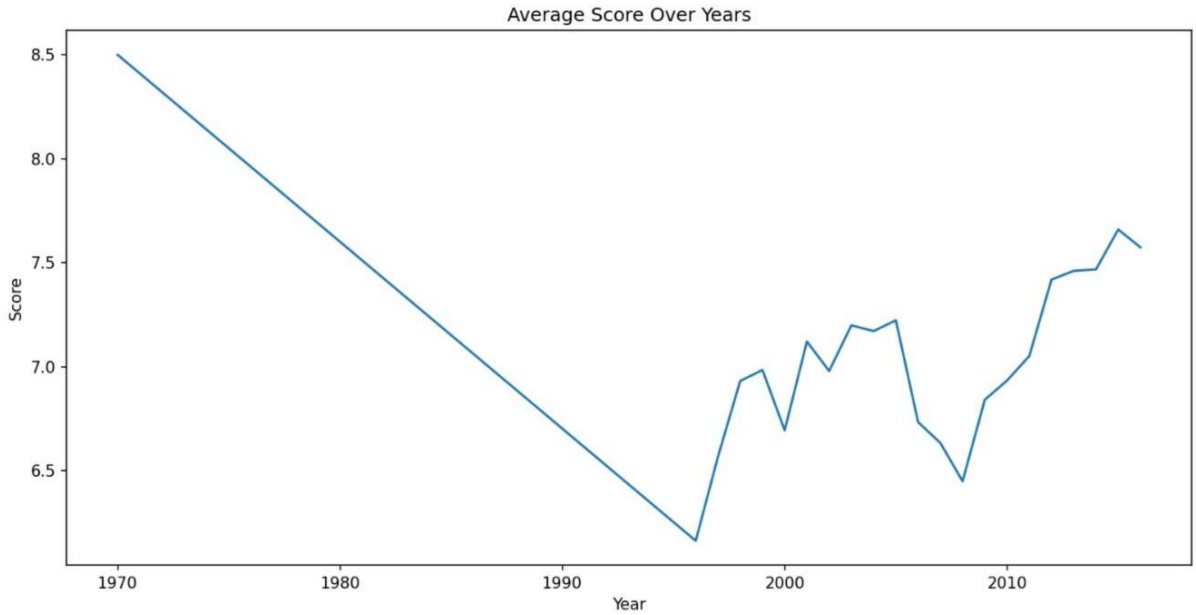
- X-axis represents years
- Y-axis represents average scores

The graph visually shows trends such as increases or decreases in scores across years.

Task	Description
1.	Group data by release_year.
2.	Calculate the average score per year using Pandas.

3.	Convert the resulting average scores and years into NumPy arrays.
4.	Plot a line graph with release_year on the X-axis and average score on the Y-axis.
5.	Add the title: "Average Game Score Over Years" and proper axis labels.
6.	Save the graph: plt.savefig("avg_score_trend.png").

Graph:



Conclusion:

The line graph reveals the trend of game scores over time.

- If the line rises → game quality is improving
- If it falls → quality is declining
- If stable → no major change

This helps understand how the gaming industry evolved in terms of review scores.

Scenario 3: Filtering + Bar Chart + Save (□ Medium)

Description:

In this scenario,

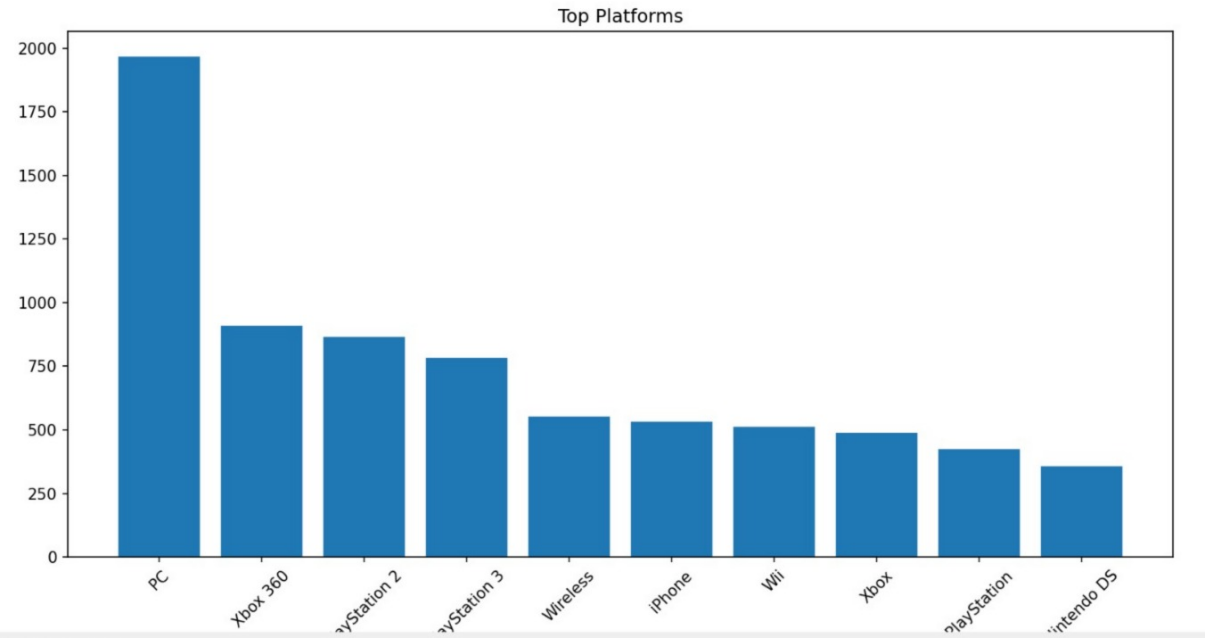
Games with score > 7 are filtered, and top 10 platforms are plotted based on count using a bar chart.

X-axis → platform names

Y-axis → number of high-rated games

Task	Description
1.	Filter the dataset to include only games where score > 7.
2.	Count the number of high-rated games per platform.
3.	Select the top 10 platforms using Pandas.
4.	Convert the platform counts into NumPy arrays.
5.	Plot a bar chart with platform on the X-axis and count of games on the Y-axis.
6.	Rotate x-axis labels for readability.
7.	Save the graph: plt.savefig("top_platforms_bar.png").

Graph:



Conclusion:

The bar chart highlights which platforms have the most high-rated games. Platforms with taller bars indicate:

- Better game quality
- Stronger performance in the gaming market

Scenario 4: Aggregation + Pie Chart + Save (□ Medium)

Description:

In this scenario,

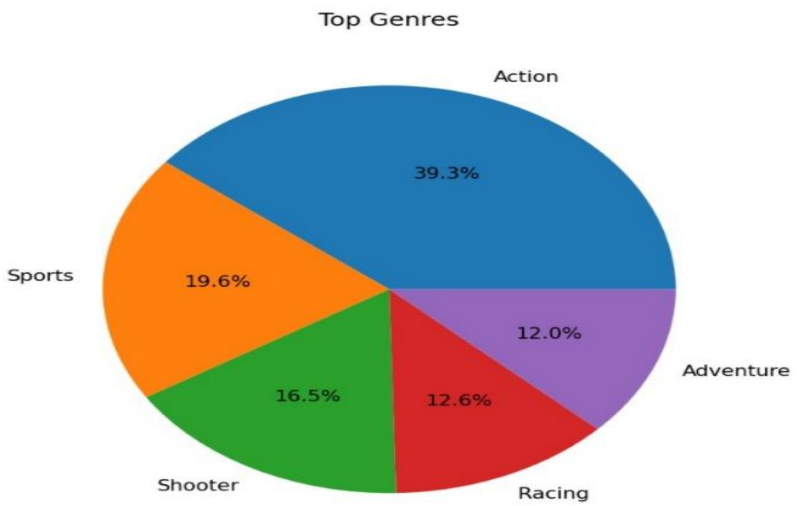
The dataset is analyzed to count how many games belong to each genre. The top 5 genres are selected to simplify visualization.

A pie chart is created to show the percentage distribution of these genres:

- Each slice represents a genre
- The size shows its proportion

Task	Description
1.	Count the total number of games per genre.
2.	Select the top 5 genres using Pandas.
3.	Prepare labels (genres) and values (counts) for the pie chart.
4.	Plot a pie chart using genre as labels and count as values.
5.	Add percentage labels using autopct (e.g., '%.1f%%').
6.	Save the graph: plt.savefig("genre_distribution.png").

Graph:



Conclusion:
The pie chart shows the most popular game genres.

- Larger slices → more games in that genre
- Smaller slices → fewer games

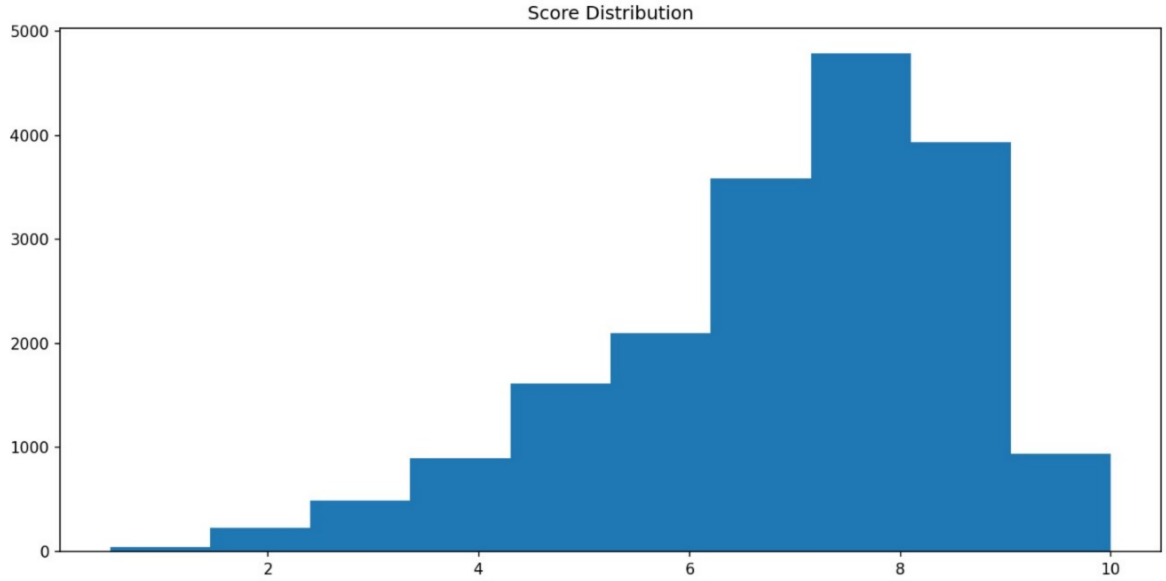
This helps understand which types of games dominate the market.

Scenario 5: Advanced Analysis + Multiple Graphs (● Hard)
Description:
In this scenario,
New features are created to improve analysis:

- Score category groups scores into:
  - Excellent (≥9)
  - Good (7–9)
  - Average (<7)
- Editor choice is converted into numeric format (1 and 0)

Task	Description
1.	Create a new column score_category: "Excellent" for score ≥ 9, "Good" for $7 \le$ score < 9, and "Average" for score < 7.
2.	Convert the editors_choice column: 'Y' $\rightarrow 1$, 'N' $\rightarrow 0$.
3.	Use NumPy's np.diff() to calculate the **yearly score growth** based on the average yearly scores from Scenario 2.
4.	Plot a **Line Graph** showing the trend of average score per release_year.
5.	Plot a **Stacked Bar Chart** showing the count of score_category per release_year.
6.	Plot a **Histogram** showing the distribution of score.
7.	Save the graphs: plt.savefig("score_trend.png"), plt.savefig("score_category_stacked.png"), plt.savefig("score_distribution.png").
8.	Identify: Which years had the highest scores, whether high scores increased over time, and if editors_choice correlates with high scores.

Graph:



Conclusion:
Shows most common score range
Helps identify if scores are mostly high, low, or balanced

Final Conclusion:
This analysis provides a complete understanding of game review data:

- Data was cleaned and prepared properly
- Trends over time were identified
- Top platforms and genres were analyzed
- Score patterns and distributions were explored

Overall, the project demonstrates how Pandas, NumPy, and Matplotlib can be used together to turn raw data into meaningful insights.

What you have learned

- How to load and clean data using Pandas
- Handling missing values and fixing data types
- Using grouping and filtering for analysis
- Converting data into NumPy arrays and using np.diff()
- Creating different graphs using Matplotlib:
  - Line chart
  - Bar chart
  - Pie chart
  - Histogram
- Saving graphs as image files
- Extracting meaningful insights from data